

is a dynamic address used for packet routing only and used by the mobile node in a foreign network. The care-of address changes every time the mobile node changes to a new sub-network. All the IP datagrams sent to the mobile node are delivered to the care-of address, not directly to the static IP address, when the mobile node is in a foreign network.

**Agents.** There are two agents: foreign agent and home agent. Known as mobility agents, these are mobile IP-aware servers or routers that know where the mobile node is actually connected.

The home agent is the home network's mobile IP agent, which is responsible for forwarding the mobile node's packets to the foreign network where the mobile node is actually connected.

The foreign agent is responsible for delivering the packets to the transient mobile node. It provides many services to the mobile node during its visit to the foreign network. The foreign agent tracks care-of address, acting as the tunnel endpoint and forwarding packets to the mobile node. The foreign agent is located in the foreign network. It is the default router for the mobile node when it is in the foreign network. The foreign agent also provides security services to the mobile node.

The home agent is located in the home network. It forwards the IP packets to the mobile node via care-of address. The home agent maintains a linked coordination between the home IP address of the mobile node and its care-of address. A tunnel is established by the home agent between itself and a reachable point (usually care-of address) for the mobile node when it is in the foreign network.

## How mobile IP works?

The routing of IP datagrams in the existing fixed Internet makes use of routers. Routers make use of the IP address of the IP datagram to route data from a source computer/node/host to the destination computer/

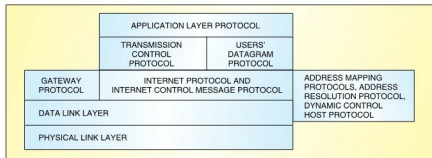


Fig. 2: TCP/IP model

node/host. The IP address comprises network ID and host ID. The network ID identifies the sub-network to which the host is connected. The host ID is the host identification number. The network is used by the routers to move datagrams from sub-network to sub-network. The final router in the target sub-network moves the packet to a particular host based on the host ID. The mobile node communicates with any other node on the Internet, whether it's mobile or not.

The entire mobile IP process is transparent. The mobile node is known to any other node by the home address of the mobile node wherever the mobile node is in physically. All packets between that node and the mobile node use the mobile node's home address regardless of whether the mobile node is in the home network or foreign network. The care-of address is used for communication only with mobility agents and is never seen by any node or host while communicating with the mobile node.

One of the important design goals of the mobile IP is to make mobility possible without major changes in the existing installed base of IPv4, particularly routing infrastructure. Mobile IP works in three steps: data transfer, getting recognition (discovery) and registration.

The data transfer phase is made of encapsulation and tunneling. It works as follows:

**Packets from any other static node to the mobile node.** 1. Any node 'A' can send a packet to mobile node 'B'. But it has to do with the home address

of mobile node 'B'. The IP datagram will carry the IP address of 'A' as the source and the home address of 'B' as the destination address.

2. The incoming packet is intercepted by the home agent in the home network. If the mobile node is within the home network, the home agent delivers the packet locally. If the mobile node is in a foreign network, the home agent forwards the packet to the foreign agent by the process of encapsulation and tunneling. The home agent encapsulates the whole incoming packet into a new IP packet with B's care-of address as the destination address, and then transmits the encapsulated packet as one IP datagram. The method of using an outer, different destination IP address is known as tunneling.

3. On receiving the encapsulated packet, the foreign agent strips off the outer IP header. This is called decapsulation. It then encapsulates the decapsulated original IP datagram in the network-level protocol data unit (for example, logical link control frame of the local-area network) and delivers the packet to 'A' in the foreign network.

**Packets from the mobile node to any other static node.** 1. Packets from mobile node 'B' to any other node 'A' are sent via the home agent when the mobile node is in the home network.

2. Packets from the mobile node to any other node are sent via the foreign agent when the mobile node is in a foreign network.

A mix of the above two techniques is used for exchange of packets be-